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DS

DarkSentinel AI

Explainable Financial Crime and Risk Intelligence Platform

AI and ML

FinTech

Cybersecurity

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Problem Statement

Why current AML systems fail at scale in the Indian financial ecosystem

The Scale: Indian financial institutions process an enormous and growing volume of UPI, NEFT, RTGS and card transactions every month. Existing rule based monitoring systems evaluate each transaction in isolation against static thresholds, with no ability to learn, adapt or explain their decisions.

1

Detection Gap

Rule based systems evaluate transactions one at a time. They cannot detect coordinated multi account schemes, layering structures or novel laundering patterns without manual rule updates.

Results in extremely high false positive alert volumes that overwhelm analyst queues, and real financial crime passes through undetected.

2

Explainability Gap

When a transaction is flagged, analysts receive only a rule code or threshold name, with no reasoning about which factors made it suspicious or how it compares to normal behaviour.

Every investigation starts from scratch, producing investigation paralysis and inconsistent decisions between analysts.

3

Access Gap

Enterprise grade AML platforms such as NICE Actimize, Oracle FCCM and SAS AML carry licensing costs that place them completely out of reach for small FinTechs, cooperative banks and microfinance institutions.

The institutions most vulnerable to financial crime end up with the weakest monitoring infrastructure.

Net Result: Large institutions are over alerted with false positives, small institutions remain under protected due to cost, and in both cases analysts make critical decisions with inadequate information.

Proposed Solution

A four layer analytical stack where each layer feeds grounded, computed data into the next

Layer 1 — Adaptive Risk Scoring Engine

- XGBoost gradient boosted classifier trained on PaySim and IEEE CIS datasets
- Over 25 engineered features covering transaction, behavioural and network signals
- Class imbalance handled with SMOTE applied to the training set only
- Isotonic Regression calibration so the output score reflects true probability
- A parallel Isolation Forest model performs unsupervised anomaly detection
- Composite score combines the supervised and anomaly scores into one result

Layer 2 — Graph Intelligence Engine

- Transactions modelled as a directed graph where accounts are nodes and transfers are edges
- Computes degree, betweenness and eigenvector centrality for every account
- Detects community structures to identify coordinated account clusters
- Finds shortest paths between an account and any known sanctioned entity
- Flags structural patterns consistent with money mule behaviour
- All graph derived signals feed back into the composite risk score

Layer 3 — SHAP Grounded Explainability

- SHAP TreeExplainer computes the exact contribution of every feature to a prediction
- Attribution is mathematically derived and unique to each transaction, not a template
- Analysts see precisely which features pushed the score up or down, and by how much
- Replaces analyst guesswork with structured, defensible evidence
- Forms the grounded input that the LLM layer is allowed to narrate
- Makes every decision traceable back to a specific numerical attribution

Layer 4 — LLM Powered Investigation Brief

- Claude API receives only the SHAP attributions, graph results and sanctions outcome
- The model is explicitly instructed to narrate findings, not invent its own risk assessment
- Cannot hallucinate a risk factor that was not first identified by the ML and graph layers
- Produces a structured brief covering risk summary, contributing factors, network concerns, sanctions status, recommended action and confidence notes
- Designed to remove repetitive manual write up work from the analyst workflow
- Output is auditable, since every sentence maps back to a computed input

System Architecture

Microservices design with async ML pipeline, role based access and tenant isolation

CLIENT LAYER

React SPA (Vite, Tailwind, React Flow, Recharts) | Role based views for Analyst, Risk Manager and Admin | Real time alerts via WebSocket

HTTPS

API GATEWAY (Node.js + Express)

- JWT RS256 Auth + RBAC
- Rate Limiter (per IP/tenant)
- Tenant ID Injection
- Audit Log Writer
- WebSocket Server
- Request Router

Ingest Service

- Validates schema
- Assigns transaction id
- Writes to MongoDB
- Pushes to score queue

Alert Mgmt Service

- Alert state machine
- Tracks SLA per alert
- SAR filing status
- Assigns to analyst

ML Scoring Service

- XGBoost + calibration
- SHAP TreeExplainer
- Isolation Forest score
- 25+ feature pipeline
- Composite score calc

Graph Service

- NetworkX directed graph
- Centrality metrics
- DBSCAN clustering
- Mule account flags

Sanctions Service

- OFAC + UN list index
- Jaro Winkler matching
- Confidence scoring
- Result caching

Report Generation Service: assembles SHAP, graph and sanctions context, calls the Claude API, returns a structured investigation brief with PDF export.

Async Design: Transactions enter a queue before scoring, so the ML service never blocks the API gateway, avoiding the bottleneck of synchronous calls under high transaction volume.

DATA LAYER MongoDB

Collection	Description
transactions	indexed: account, time, score
alerts	state machine, SAR status
accounts	indexed: account, tenant
audit_logs	immutable, long retention
sanctions_cache	OFAC + UN, timed cache
model_metrics	performance over time
users	hashed creds, roles, tenant

Technology Stack

Every component selected for correctness, scalability and auditability, not convenience

Frontend

- React.js with Vite
- Tailwind CSS
- React Flow for graph visualisation
- Recharts for analytics and SHAP waterfall charts
- Role based dashboard views
- WebSocket client for live alerts

API Gateway

- Node.js with Express
- JWT RS256 with refresh rotation
- Role based access control middleware
- Helmet.js security headers
- Per IP and per tenant rate limiting
- Tenant isolation enforcement

ML Service

- Python with FastAPI
- XGBoost fraud classifier
- Isotonic Regression calibration
- SHAP TreeExplainer
- Isolation Forest anomaly detection
- NetworkX graph analytics

Data Layer

- MongoDB with seven collections
- Indexes on account, time and risk score
- Time to live indexes for retention policy
- PaySim and IEEE CIS training data
- OFAC and UN sanctions index
- Model performance metrics store

LLM and Reports

- Claude API for narrative generation
- Strictly grounded prompting design
- Structured multi section brief output
- PDF export of investigation brief
- Report linked to alert in MongoDB
- Every call logged in audit trail

Datasets and Training Approach

Each dataset is mapped to a specific model with a documented rationale

PaySim

Approximately 6.3 million transactions

Role: Primary training data for the XGBoost fraud classifier

Mobile money simulation with transaction types CASH IN, CASH OUT, TRANSFER and PAYMENT that closely mirror UPI and NEFT patterns. Also used to build the sender to receiver transaction graph.

IEEE CIS Fraud Detection

Large scale, hundreds of features

Role: Secondary classifier and source of feature engineering ideas

Industry grade e commerce fraud dataset. Its velocity, device and email domain feature patterns are adapted into the engineered feature pipeline.

ULB Credit Card Fraud

PCA transformed features

Role: Used only to train the Isolation Forest anomaly model on normal transactions

The extreme class imbalance makes it ideal for unsupervised anomaly detection trained on the majority class. It is deliberately not used for the supervised classifier because PCA transformed features would make SHAP attributions meaningless.

OFAC SDN List

Large entity and alias index

Role: Primary sanctions screening source

United States Treasury list with structured alias fields. Indexed in memory and matched using Jaro Winkler fuzzy similarity across all aliases, not just primary names.

UN Consolidated Sanctions List

International coverage

Role: Supplements OFAC for non US designations

Combined with OFAC to build a single merged sanctions index covering both United States and international designations.

FATF High Risk Jurisdictions

Country level risk tiers

Role: Country risk feature in the scoring model

Countries on the high risk and increased monitoring lists contribute a country risk adjustment to the composite score, reflecting real regulatory guidance.

Expected Impact

Directional outcomes the design targets, to be measured once the system is built and evaluated

Note: *These are design goals based on the architecture, not measured results. Actual figures will be produced after training, calibration and evaluation against the baseline comparison described below.*

Time

Faster Investigation

Each alert is intended to arrive with a pre built investigation brief, a SHAP attribution chart and graph context, replacing the current practice of starting every investigation from a blank page.

Filter

Fewer False Positives

By replacing static thresholds with a calibrated probability score plus anomaly detection and network context, the system aims to reduce the proportion of alerts that turn out to be false positives compared with a static rule based baseline.

Scale

Higher Analyst Throughput

Structured, auto generated reports are intended to let each analyst review more alerts per day than with fully manual investigation, since the repetitive write up work is automated.

EVALUATION FRAMEWORK (to be populated after training):

System	Detection Method	Explainability	Expected Direction
Static rule based threshold	Fixed rules per field	Rule name only	Reference baseline
Logistic regression on raw data	Naive linear model on imbalanced data	Coefficient weights, not per case	Expected to underperform due to imbalance
DarkSentinel calibrated XGBoost	Calibrated probability plus anomaly and graph signals	Per transaction SHAP attribution	Target: improved precision recall trade off and lower false positive rate, measured after evaluation

Project Roadmap

A phased build plan moving from data and models through to a demo ready platform

Phase 1

Data and Model Foundation

- Download and explore PaySim and supporting datasets
- Build the feature engineering pipeline (25+ features)
- Time based train, validation and test split to avoid leakage
- Train XGBoost with SMOTE and calibrate with Isotonic Regression
- Train Isolation Forest on normal transactions only
- Integrate SHAP TreeExplainer and verify attributions

Phase 2

Core Services

- Scaffold the monorepo and docker compose setup
- Build the Ingest Service with schema validation
- Build the ML Scoring Service exposing a score endpoint
- Build the Alert Management Service with its state machine
- Set up the seven MongoDB collections with indexing

Phase 3

Graph and Sanctions Intelligence

- Build the NetworkX based Graph Service
- Implement centrality metrics and DBSCAN clustering
- Parse OFAC and UN sanctions lists into a merged index
- Implement Jaro Winkler fuzzy matching with caching
- Wire graph and sanctions outputs into the composite score

Phase 4

Gateway, Dashboard and Security

- Build the Node.js API Gateway with JWT and RBAC
- Implement rate limiting and tenant isolation
- Build the React dashboard for all three personas
- Add the SHAP waterfall chart and graph visualisation
- Apply Helmet, input validation and audit logging

Phase 5

LLM Reports and Demo Preparation

- Integrate the Claude API with a strictly grounded prompt
- Generate structured, multi section investigation briefs
- Add PDF export of the investigation brief
- Load a demo dataset and prepare showcase scenarios
- Record a walkthrough and prepare documentation